

H3 — ML + Embedded Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: the team understands the model file format and the export guards.

Steps: 1. Read research/tinystories/export.py and sample.py. 2. Write down what these mean: PLE format, model.bin, golden.txt, golden.npz, tied head, tokenizer hash. 3. Write the 1-page export explainer. Answer in it: why does export refuse a wrong tokenizer?

Deliverable: the export explainer as a markdown file in the club repo.

Verify: you can explain the `--allow-unverified-tokenizer` flag from memory.

Learning this week

Tokenization. Read how the tokenizer maps text to ids. Explain why a 32768-token vocab needs a big table.

Resources

- export.py: <https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/export.py>
- sample.py: <https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/sample.py>
- research README: <https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/README.md>

Done when

- [] Export explainer exists in the club repo
- [] You can explain the tokenizer hash check from memory
- [] You can say what the project does in one sentence